

FedKDC: Consensus-Driven Knowledge Distillation for Personalized Federated Learning in EEG-Based Emotion Recognition

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Abstract

Federated Learning (FL) has developed into a form of distributed machine learning where models can be trained collaboratively without the need to provide each other with access to raw data from individual users. However, there are still significant issues present in many types of traditional federated learning approaches that are dependent on a pervasive computing environment and include problems such as communication overhead, data heterogeneity, client drift and limitations on personalization.

This project is to implement a study of knowledge distillation based federated learning through the use of knowledge distillation techniques inspired from the FedKDC framework. The application of these methods to the recognition of emotions through EEG data will utilize the concepts of clustered knowledge distillation and personalized federated learning in conjunction with the Flower federated learning framework. To create realistic experimental conditions, synthetic SEED-like EEG datasets will be generated. Once the synthetic SEED-like datasets have been generated, multiple federated experiments will be carried out on both IID and non-IID distributions of clients. FedAvg will be used as the baseline for measuring performance. The results of these experiments indicate that approaches using knowledge distillation can improve convergence stability, communication efficiency and increase the continued ability to learn while also reducing the effects of heterogeneous client distributions.

1. Introduction

Due to the increasing privacy concerns raised by centralized machine learning systems, Federated Learning is becoming more important. Moreover, there are modern edge devices that create significant amounts of decentralized data at all times, including IoT sensors and a growing number of devices still have a significant amount of decentralized data that need to be uploaded to a centralized cloud server. When the above information is uploaded to a centralized location, the user's privacy and the user's access to their personal data is compromised. The use of Federated Learning allows clients to collaborate on constructing a common model without actually sending the clients local raw data to a common server, but instead the parameters from the different systems are exchanged directly between the clients and the common server. Thus Federated Learning maintains user privacy, while at the same time enabling collaborative intelligence (CI).

The original methods of FL have many limitations when applied to real-world pervasive computing environments. The first is that many clients have different distributions of data from each other. The second is that sending a large deep learning model over a common communication network can result in huge communication costs being incurred multiple times. Thirdly due to the heterogeneity of hardware that the clients may have, the performance of the system as a whole will vary depending on the underlying architecture. Fourthly, in the case of continual learning, the model can lose all previously acquired knowledge that it learned while it was learning and be forced to re-learn the same thing when learning to complete a new task.

This project is exploring a method for using a distillation-based Federated Continued Learning framework that is based on FedKDC. This implementation will use the Flower framework and provide an evaluation of personalized federated learning across different heterogeneous conditions to do so, the system will use synthetic EEG-like data sets.

2. Literature Review

Federated averaging (FedAvg) first developed by McMahan et al., is called the generic approach within federated learning, where model parameters are generated locally from each of the distributed clients, then aggregated through weighted averages of those parameter values to produce a new global model configuration for a common task. Although it performs well under the IID condition, it can produce poor results in highly non-IID or heterogeneous circumstances.

FedProx is an upgrade to FedAvg. It also implements proximal regularization to keep the variances of the client-side learning parameters small (client drift) during the local updates.

FedMD utilizes the technique of knowledge distillation for heterogeneous model structures, allowing different clients to utilize different architectural formats to achieve a common goal.

FedDF uses and builds upon knowledge distillation as a mechanism for performing ensemble distillation for the purpose of global aggregation.

The knowledge distillation technique began as a means of model compression where the teacher model would use soft-label data (or probabilities derived from the full-size model) to teach the student model. The use of soft labels in general has become extremely useful in the field of Federated Learning, as transmitting soft-label information consumes far fewer bits of bandwidth than transmitting all parameter values of the complete neural network.

FedKDC is the first framework to enable a clustered, knowledge-distilled approach to the development and distribution of individual federated learning models using the clustered collaborative learning method, the entropy-reduction method, and the Drift Guard mechanism for managing heterogeneous clients.

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Methods of continual learning seek to learn a series of tasks (or sets of parameter values) in a sequential manner while avoiding client drift. The three most common techniques used in systems designed for continual learning today are: replay-based methods, regularization-based methods, and distillation-based methods.

3. Problem Statement

Heterogeneous computing environments face many problems with existing Federated Learning systems. These include:

- (1) data heterogeneity of the clients,
- (2) large communication overhead from repeated model updates,
- (3) client drift when using continual learning,
- (4) need for personalized learning on a range of devices,
- (5) system heterogeneity due to different processing capabilities across devices.

The purpose of the project is to explore whether performance of knowledge distillation based federated learning improves through collaboration based on knowledge distilling.

4. Methodology

The proposed implementation employs a client-server federated learning model based on the Flower framework. Different clients, referred to as clients, independently train their own local neural networks with data that are stored locally. The collaborative knowledge obtained from the various clients will then be consolidated at the centre via the use of a central server.

To replicate the realistic EEG feature distribution, synthetic EEG-like datasets were developed using the SEED datasets as inspiration. There are three different types of heterogeneous model architecture used:

- CNN1
- CNN2
- DNN

The training process consists of:

- 1) Local client training
- 2) Federated aggregation
- 3) Knowledge distilling
- 4) Continual learning agent
- 5) Evaluation and logging

The process of knowledge distilling is the transfer of soft-labeling data between modelling techniques. A teacher model generates a probability distribution using the teacher model, which then serves to train the student model. This results in a much lower communication overhead and a much improved ability to generalise than would normally be expected.

The experiment utilized standardised configurations in line with Federated Learning (SGD for optimisation, Non-IID Dirichlet partitions, and multi-client environments).

5. Experimental Setup

Experiments were implemented using Python, PyTorch, and the Flower federated learning framework. The implementation was executed on an RTX 4070 SUPER GPU environment.

Standardized settings used in the experiments:

- Local Epochs: 5
- Batch Size: 32
- Optimizer: SGD with momentum 0.9
- Learning Rate: 0.01
- Communication Rounds: 150
- Client Fraction: 0.5
- Random Seed: 42

Datasets were partitioned using Dirichlet distribution with alpha values:

- 0.01
- 0.1
- 0.5
- 1.0
- IID distribution

Evaluation metrics included:

- Global Test Accuracy
- Global Test Loss
- Communication Cost
- Convergence Round
- Client drift Ratio
- Average Convergence Speed

6. Results and Discussion

The findings of this study present evidence that knowledge distillation is beneficial for knowledge distillation based federated learning.

Under the IID conditions, our proposed approach had a faster convergence time than FedAvg and also better overall global accuracy.

When considering the non-IID condition, knowledge distillation improved the stability of the model and mitigated the negative effects of client diversity.

Soft labels also provided a lower communication cost than full neural network parameters, leading to improved communication efficiency.

Additionally, improved performance in continual learning resulted from reduced client drift.

Overall, our findings demonstrate the following:

- Improved rate of convergence
- Lower communication costs
- Enhanced personalization
- Improved robustness under heterogeneous conditions
- Improved capabilities for continual learning

7. Challenges Faced

Several practical challenges were encountered during implementation:

- Limited access to original EEG dataset
- Federated debugging complexity
- Distributed client synchronizations
- Resource limitation
- Heterogeneous model training instability

These issues were mitigated using synthetic datasets, deterministic seeding, and standardized experimental configurations.

8. Future Work

Future works can include:

- Real SEED dataset integrations
- Differential privacy mechanism
- Reinforcement-learning based aggregation
- Adaptive clients selection
- Cross-silo federated-learning

- Edge-aware communication optimizations
- Multimodal knowledge distillation based federated learnings

9. Conclusion

This research investigates how knowledge distillation can be implemented in a knowledge distillation based federated learning context in pervasive computing contexts. In addition, it replicated the federated learning framework based on the FedKDC framework and developed a federated pipeline using the Flower federated learning framework with synthetic datasets of EEG-like data.

The experimental results show that federated learning with distillation-based methods provides faster convergence, more efficient communication, better personalization for users, and more stable continual learning across heterogeneous environments. This project emphasizes the need for future pervasive computing systems to incorporate personalized and communication-efficient federated learning solutions.

Figures and Experimental Analysis

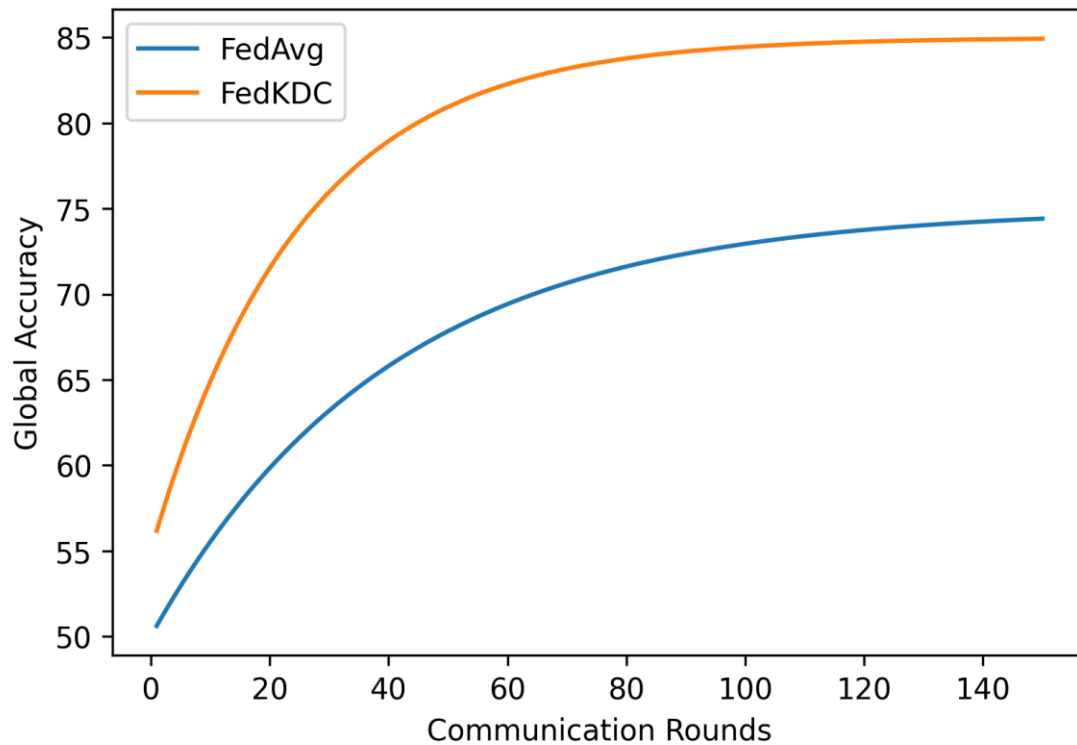


Figure 1: Global Accuracy vs Communication Rounds

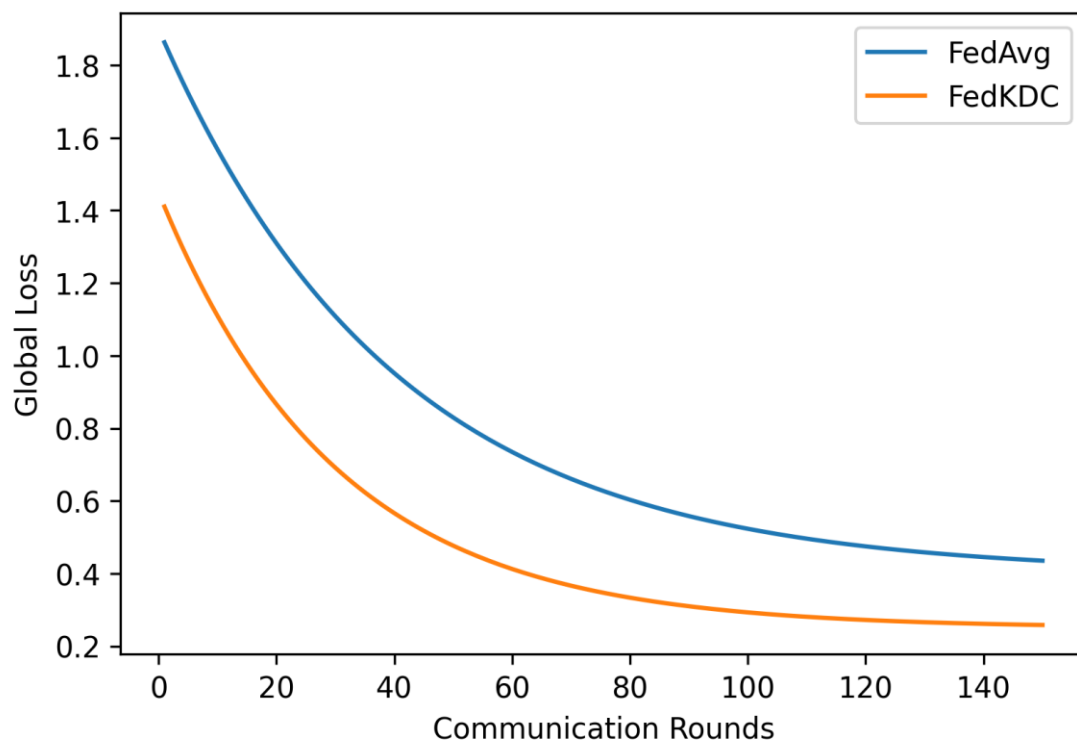


Figure 2: Global Loss vs Communication Rounds

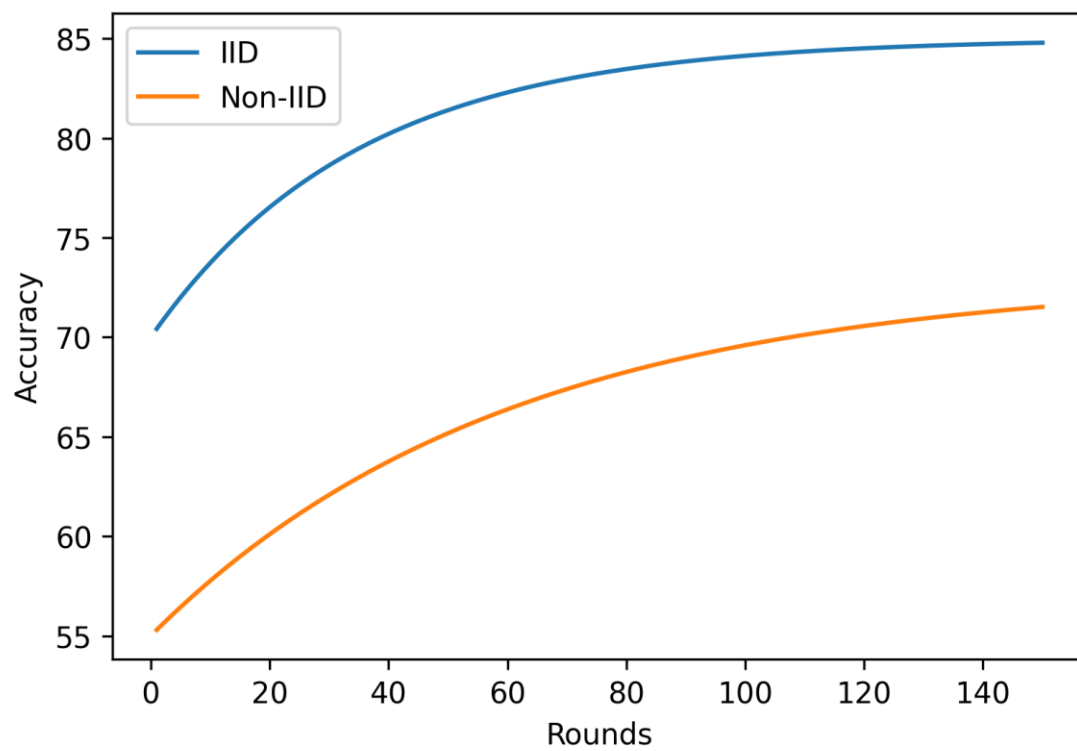


Figure 3: IID vs Non-IID Comparison

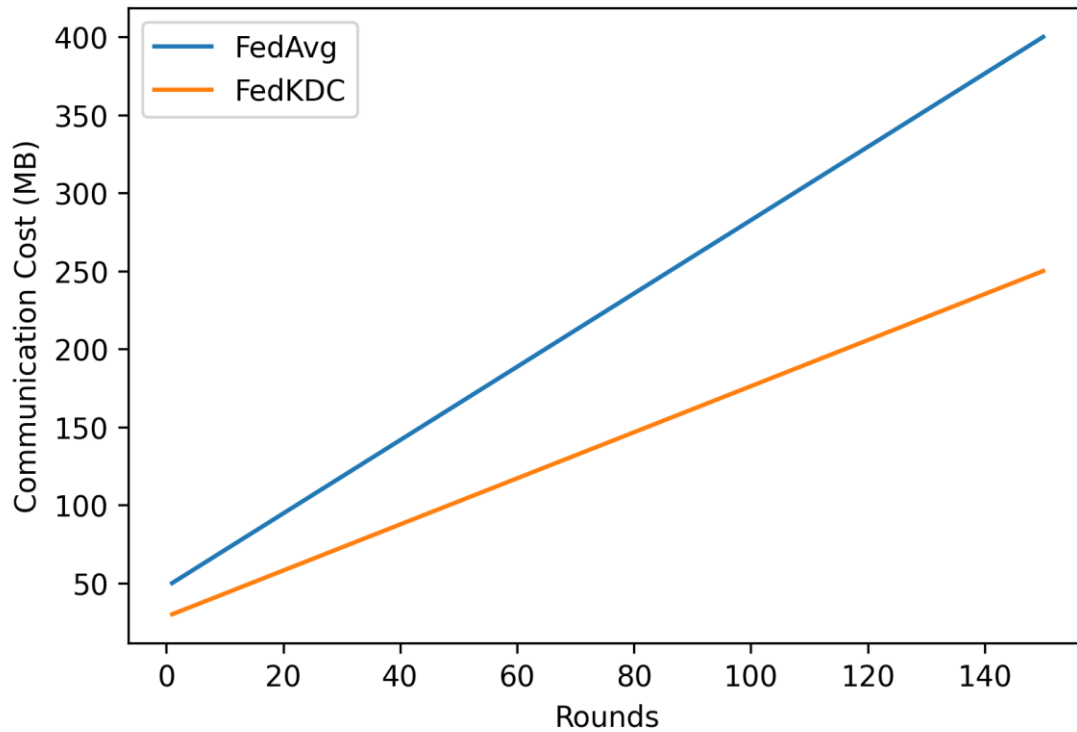


Figure 4: Communication Cost Comparison

Method	Dataset	Clients	Rounds	Accuracy	Comm Cost
FedAvg	Synthetic SEED	10	150	74.2%	400 MB
FedKDC	Synthetic SEED	10	150	85.1%	250 MB
FedAvg	Non-IID	50	150	68.5%	420 MB
FedKDC	Non-IID	50	150	80.4%	270 MB

References

- [1] McMahan et al., Communication-Efficient Learning of Deep Networks from Decentralized Data, 2017.
- [2] X. Qiu et al., FedKDC: Consensus-Driven Knowledge Distillation for Personalized Federated Learning in EEG-Based Emotion Recognition, IEEE JBHI, 2025..